Draft Literature Review

Table of Contents

Draft Literature Review 1
1. Core Concepts 2
1.1. Motivation 2
1.2. Gamification 2
2. Problem Domain 3
2.1. Existing Sources 3
2.2. Issues ..... **Error! Bookmark not defined.**
3. Technologies 4
4. References 5

An excellent literature review will provide a clear reference point for the project. It will examine the state of the art in the relevant field, show where similar work has already been done and make clear where there is scope for further work. Discussion will demonstrate a breadth of awareness and solid understanding of relevant principles and techniques. Full source references will be made to background material on all significant issues concerned, including traditional and formal academic sources. A high-class report will be further strengthened by a critical appraisal of the problem domain.

1. Core Concepts

1.1. Motivation

The primary focus of this project is to create an application that supports students in their studies and encourages them to work independently and manage their time effectively. To increase the user's motivation to study, first we must understand how best to self-motivate, and to include those methods into the project.

In the conference paper titled 'Exploring Goal-setting, Rewards, Self-monitoring, and Sharing to Motivate Physical Activity' (Munson & Consolvo, 2012), the researchers posited that a specialised application could be implemented to improve the self-motivation of users. The paper suggests that users are not motivated by awards given by motivation-based applications as they are already aware of whether they have or have not achieved their desired goal. The users described in the paper also had mixed feelings on the ability to share their goals on social media, some stating that they felt uncomfortable with the public nature of their progress and were afraid of judgement from their peers. Despite these concepts having been laid out with Gamification, the exploratory field study suggest that these features become a detriment to the desired effect of the application. As such, the proposed application will focus more on rewarding motivated activity with in-app currency instead of a more abstract reward and motivating users without utilising sharing or peer-based encouragement, instead focusing on motivating without shaming or judging the user to create a friendlier and less stressful environment.

The 'Ace your Self-Study' app (*Ace your Self-Study app | Educational Innovation | Erasmus University Rotterdam*, n.d.) is a tool developed to help students cultivate their self-regulated (Zimmerman, 2002) learning and encourage students to reflect and improve on their processes while studying. The principles of the application are to promote critical self-analysis and to guide users to better habits without removing the user's autonomy. The study 'Exploring Students' Use of a Mobile Application to Support Their Self-Regulated Learning Processes' (Baars et al., 2022) demonstrates the benefits of self-regulated learning when paired with a mobile application such as the one proposed in this project.

1.2. Gamification

Gamification is the concept of adding traditional game features, such as levelling, awards such as badges, leaderboards, and goals, to an existing software or tool to engage the user and foster a deeper understanding of the material. The interactive aspects have shown to improve academic achievement and further cognitive development, as well as their motivation to continue a given task. When such techniques were studied in the paper 'Gamification Strategy to Support Self-Directed Learning in an Online Learning Environment' (Palaniappan & Md Noor, 2022) the participants were said to 'have significantly improved' their academic performance. The paper also found that participants were engaging in more self-directed learning after gamification was included and that gamification 'gamification strategy used in the online learning environment shows a positive influence' on both the participant and the content being taught.

2. Problem Domain

2.1. Existing Sources

'Forest' (*Forest - Stay focused, be present*, n.d.), is an application designed to promote focus in users by preventing users from accessing other applications while they are supposed to be studying. As they study a timer counts down until they can use their phone again, and as the timer continues a plant or tree grows. The gamification techniques included are designed to convey consequences for losing focus, and guiding the user towards smaller, more achievable goals. The application also includes some in-app currency which can be exchanged for cosmetic changes to what the user 'grows', allowing users to customise their experience and feel more connected with their productivity.

2.2. Confronted Issues

During the COVID-19 pandemic, educational bodies turned to online learning. As online learning had not been attempted at such scale before, many institutions were not sufficiently equipped to adapt their methodologies for teaching and as such resulted in higher stress and lack of motivation from students (Lemay et al., 2021). While online learning has become a larger part of academia in a post-covid world, there has been minor change in the approach of how to motivate students when not in the classroom.

According to the research paper 'Mental Health, Academic Self-Efficacy and Study Progress Among College Students – The SHoT Study, Norway' (Grøtan et al., 2019) mental disorders such as depression and anxiety are common in university students, leading to decreased work output and impaired memory. Due to their lack of productivity, students with these disorders have shown to become lethargic, and have 'lowered mood[s]', which also result in a decrease in productivity. This spiral of shame and fear can often leave students feeling paralysed in their decisions and worsen. Creating an application that would aim to encourage users and not shame them would allow students with mental disorders to break this cycle and change their behaviours more favourably.

3. Technologies

(*Firebase | Google's Mobile and Web App Development Platform*, n.d.)

(*Docker: Accelerated Container Application Development*, n.d.)

(*React Native · Learn once, write anywhere*, n.d.)

(*Windows Server 2025 | Microsoft Evaluation Center*, n.d.)

4. References

1. *Ace your Self-Study app | Educational Innovation | Erasmus University Rotterdam* (n.d.). https://www.eur.nl/en/educational-innovation/news/ace-your-self-study-app [Accessed 12 Jan 2026].
2. Baars, M., Khare, S. & Ridderstap, L. (2022) Exploring Students' Use of a Mobile Application to Support Their Self-Regulated Learning Processes. *Frontiers in Psychology*, 13, p. 793002. Available online: https://doi.org/10.3389/fpsyg.2022.793002.
3. Deterding, S., Dixon, D., Khaled, R. & Nacke, L. (2011) From game design elements to gamefulness: defining 'gamification'. In *Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments*. *MindTrek '11: Academic MindTrek 2011*, ACM, p. 9–15. Available online: https://doi.org/10.1145/2181037.2181040.
4. *Docker: Accelerated Container Application Development* (n.d.). https://www.docker.com/ [Accessed 13 Jan 2026].
5. *Firebase | Google's Mobile and Web App Development Platform* (n.d.). https://firebase.google.com/ [Accessed 13 Jan 2026].
6. *Forest - Stay focused, be present* (n.d.). https://www.forestapp.cc/ [Accessed 12 Jan 2026].
7. Grøtan, K., Sund, E.R. & Bjerkeset, O. (2019) Mental Health, Academic Self-Efficacy and Study Progress Among College Students – The SHoT Study, Norway. *Frontiers in Psychology*, 10, p. 45. Available online: https://doi.org/10.3389/fpsyg.2019.00045.
8. Lemay, D.J., Bazelais, P. & Doleck, T. (2021) Transition to online learning during the COVID-19 pandemic. *Computers in Human Behavior Reports*, 4, p. 100130. Available online: https://doi.org/10.1016/j.chbr.2021.100130.
9. Munson, S. & Consolvo, S. (2012) Exploring Goal-setting, Rewards, Self-monitoring, and Sharing to Motivate Physical Activity. In *Proceedings of the 6th International Conference on Pervasive Computing Technologies for Healthcare*. *6th International Conference on Pervasive Computing Technologies for Healthcare*, IEEE. Available online: https://doi.org/10.4108/icst.pervasivehealth.2012.248691.
10. Palaniappan, K. & Md Noor, N. (2022) Gamification Strategy to Support Self-Directed Learning in an Online Learning Environment. *International Journal of Emerging Technologies in Learning (iJET)*, 17(03), p. 104–116. Available online: https://doi.org/10.3991/ijet.v17i03.27489.
11. *React Native · Learn once, write anywhere* (n.d.). https://reactnative.dev/ [Accessed 13 Jan 2026].
12. Sierra Nevada College. Department of Humanities, Psychology Program., Ruuymaekers, A. & Frederick, C. (2018) Ascending versus Descending Timers: Stress and Motivation. *Nevada State Undergraduate Research Journal*, 4(1), p. 25–30. Available online: https://doi.org/10.15629/6.7.8.7.5_4-1_S-2018_3.
13. *Windows Server 2025 | Microsoft Evaluation Center* (n.d.). https://www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2025 [Accessed 13 Jan 2026].
14. Zimmerman, B.J. (2002) Becoming a Self-Regulated Learner: An Overview. *Theory Into Practice*, 41(2), p. 64–70. Available online: https://doi.org/10.1207/s15430421tip4102_2.